Code : 303202

BCA 2nd Semester Exam., 2014

MATHEMATICS (NUMERICAL TECHNIQUES)

Time : 3 hours *Full Marks* : 60

Instructions :

(i) All questions carry equal marks.

(ii) There are **SEVEN** *questions in this paper.*

(iii) Attempt **FIVE** *questions in all.*

(iv) Question Nos. **1** *and* **2** *are compulsory.*

1. Choose the correct options (any *six*) :

(a) If three approximate values of the number $\frac{1}{3}$ are given as 0·30, 0·33 and 0·34, then which of these three is the best approximation?

(i) 0·30

(ii) 0·33

(iii) 0·34

(b) If a function f is continuous between a and b and $f(a)$ and $f(b)$ are of opposite signs, then the number of roots of $f(x)=0$ lying between a and b is

(i) exactly one

(ii) at least one

(iii) exactly two

(iv) at least two

(c) If $y_0, y_1, y_2, \cdots, y_n$ denote a set of values of y, then which of the following represents the second forward difference $\Delta^2 y_0$?

(i) $y_2 + 2y_1 - y_0$

(ii) $y_2 - 2y_1 + y_0$

(iii) $y_2 - 2y_1 - y_0$

(iv) None of the above

(d) Which of the following is correct, where the operators have their usual meanings?

(i) $\delta = E^{\frac{1}{2}} + E^{-\frac{1}{2}}$

(ii) $\mu = E^{\frac{1}{2}} + E^{-\frac{1}{2}}$

(iii) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$

(iv) $\mu = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$

(e) The $(n+1)$th divided difference of a polynomial of degree n is

(i) 0

(ii) 1

(iii) n

(iv) None of the above

(f) Which of the following formulae should be used, if interpolation is required near the beginning of a set of tabular values?

(i) Newton's forward interpolation formula

(ii) Newton's backward interpolation formula

(iii) Stirling's formula

(iv) Bessel's formula

(g) While using Simpson's $\frac{1}{3}$ rule, the number of equal subintervals, into which the given interval must be divided, should be a multiple of

(i) 2

(ii) 3

(iii) 4

(iv) 6

(h) Gauss elimination method for solving a system of linear equations reduces the system to an equivalent system which is

(i) diagonal

(ii) upper triangular

(iii) lower triangular

(iv) None of the above

(i) The system

$$2x + y = 2$$
$$2x + 1 \cdot 01y = 2 \cdot 01$$

is

(i) not consistent

(ii) ill-conditioned

(iii) well-conditioned

(iv) neither ill-conditioned nor well-conditioned

(j) If the interval of differencing is 1, then the value of $\Delta\left(\frac{1}{x}\right)$ is

(i) $\frac{1}{x}$

(ii) $\frac{1}{x+1}$

(iii) $\frac{1}{x(x+1)}$

(iv) $-\frac{1}{x(x+1)}$

2. Answer any *three* of the following :

(a) Define absolute, relative and percentage errors. Evaluate the sum $\sqrt{3}+\sqrt{5}+\sqrt{7}$ to four significant digits.

(b) Define the operators Δ and ∇. Prove that

$$\Delta(y_k^2) = (y_k + y_{k+1})\Delta y_k$$

(c) Derive the Newton-Raphson formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

(d) Prove that the third divided difference of the function $f(x) = \frac{1}{x}$ with arguments p, q, r, s is

$$-\frac{1}{pqrs}$$

(e) Given that $\frac{dy}{dx} - 1 = xy$ and $y(0) = 1$. Obtain the Taylor series for $y(x)$.

3. Find a root between 0 and 1 of the equation $x^3 + x - 1 = 0$ by the method of bisection correct to three decimal places.

4. Applying Lagrange's interpolation formula, find a cubic polynomial which approximates the following data :

x	-2	-1	2	3
$y(x)$	-12	-8	3	5

5. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by—

(a) Simpson's $\frac{1}{3}$ rule;

(b) Simpson's $\frac{3}{8}$ rule;

using six subintervals.

6. Solve the system of equations

$$5x - 2y + z = 4$$
$$7x + y - 5z = 8$$
$$3x + 7y + 4z = 10$$

by Gauss elimination method.

7. Use Runge-Kutta method to find $y(0 \cdot 1)$, given that

$$\frac{dy}{dx} = \frac{1}{x+y}, \quad y(0) = 1$$

★★★

BCA 2nd SEMESTER EXAM., 2015
MATHEMATICS CODE- 303202

Time: 3 hours Full Marks: 60

Instructions:

i. The Marks are indicated in the right -hand margin.
ii. There are ***SEVEN*** questions in this paper.
iii. Attempts ***FIVE*** question in all.
iv. Question Nos. ***1*** and ***2*** are compulsory.

1. Choose the correct option (any six). 2*6=12

(a) The rounded value of 57.275 to two decimal place is

(I) 57.27 (II) 57.28

(III) 57.26 (IV) None of the above

(b) A non-zero polynomial $f(x)$ of degree 3 has roots at x = 1, x = 2 and x = 3.

Which of the following is true?

(I) $f(0) + f(4) < 0$ (II) $f(0)f(4) > 0$

(III) $f(0) + f(4) > 0$ (IV) None of the above

(c) The function $f(x) = 1 + \cos x - 5x^2$ is an example of

(I) Polynomial (II) Transcendental

(III) Both (I) and (II) (IV) None of the above

(d) Newton-Raphson method is based on Taylor's series but

(I) neglecting 3rd and heigher order derivatives

(II) accepting up to 3rd order derivatives

(III) accepting up to 4th order derivatives

(IV) neglecting 2nd order and higher order derivatives

(e) If $y_0, y_1, \ldots y_n$ denote a set of values of y, then Δy_0 is

(I) (E + 1) y_0 (II) E y_0

(III) (E - 1) y_0 (IV) none of the above

(f) Simpson's $\frac{1}{3}rd$ rule requires the division of the entire into

(I) odd number of sub-intervals of equal length

(II) even number if sub-intervals of equal length

(III) odd or even numbers of sub-intervals of equal length

(IV) None of the above

(g) Back-substitution is used in

(I) Jacobi method (II) Gauss elimination method

(III) Gauss-Jordan method (IV) none of the above

(h) Convergence in the Gauss-Seidel method is __________as fast as Gauss-Jacobi method

(I) same (II) twice

(III) thrice (IV) None of the above

(i) Lagrange polynomial of degree two passes

(I) one point (II) two points

(III) three points (IV) none of the above

(j) Error in Simpson's $\frac{3}{8}rd$ rule is __________compare to Simpson's $\frac{1}{3}rd$ rule.

(I) small (II) negligible

(III) zero (IV) large

2. Answer any three of the following: 4*3=12

(a) State the principle used in Gauss-elimination method.

(b) Solve $x + 2y = 5$

$2x + y = 4$

using Gauss-Jordan method.

(c) State Newton's algorithms for finding square root of ***N.***

(d) If the matrix A is such that $A = \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} \begin{pmatrix} 1 & 9 & 5 \end{pmatrix}$

then find det(A).

(e) Find $\Delta \sin x$.

Answer any three of the following: 12*3=36

3. Find the missing term of the table given below :

X :	1	2	3	4	5	6	7
Y :	2	4	8	--	32	64	128

4. The velocity V km/min of a motorbike that starts from rest is given below :

t :	2	4	6	8	10	12	14	16	18	20
V :	10	18	25	29	32	20	11	5	2	0

Find the approximate distance covered in twenty minutes using Simpson's $\frac{1}{3}rd$ rule.

5. Use Jacobi method to find the solution of the following set of linear equations :

$5x_1 - 2x_2 + 3x_3 = -1$

$-3x_1 + 9x_2 + x_3 = 2$

$2x_1 - x_2 - 7x_3 = 3$

6. Find $\Delta^n (e^{ax+b})$.

7. Describe Lagrange's interpolation formula. Consider the following x_i's :

i	:	0	1	2
x_i	:	1	3	-2

Find $L_0(x)$, $L_1(x)$ and $L_2(x)$

* * *

Code : 303202

BCA 2nd Semester Exam., 2016

MATHEMATICS

Time : 3 hours *Full Marks* : 60

Instructions :

(i) *All questions carry equal marks.*

(ii) *There are **SEVEN** questions in this paper.*

(iii) *Attempt **FIVE** questions in all.*

(iv) *Question Nos. **1** and **2** are compulsory.*

1. Choose the correct answer (any *six*) :

(a) If the function $f(x) = x\sin x$ satisfies

$$f''(x) + f(x) + t\cos x = 0$$

then the value of t is

(i) 1

(ii) 2

(iii) -2

(iv) None of the above

(b) A non-zero polynomial $f(x)$ of degree 3 has roots at $x = 1$, $x = 2$ and $x = 3$. Which of the following is true?

(i) $f(0)\,f(4) < 0$

(ii) $f(0)\,f(4) > 0$

(iii) $f(0) + f(4) > 0$

(iv) $f(0) + f(4) < 0$

(c) The real root of the equation $\cos x - 3x + 1 = 0$ correct up to 4 decimal places by the method of iteration is

(i) 0·8071

(ii) 0·6071

(iii) 0·7071

(iv) 0·9071

(d) The process of estimating the value of dependent variable at an intermediate value is called

(i) interpolation

(ii) extrapolation

(iii) estimation

(iv) dependency

(e) Simpson's $\frac{1}{3}$rd rule is obtained by taking $n =$ —— in the general quadrature formula.

(i) 1 *(ii)* 2

(iii) 3 *(iv)* 4

(f) Gauss-Jacobi method is a/an _____ method.

(i) non-iteration

(ii) iteration

(iii) algebraic

(iv) None of the above

(g) Convergence in the Gauss-Seidel method is _____ as fast as Gauss-Jacobi method.

(i) thrice

(ii) same

(iii) twice

(iv) None of the above

(h) Error in Simpson's $\frac{3}{8}$th rule is _____ compared to Simpson's $\frac{1}{3}$rd rule.

(i) small

(ii) negligible

(iii) zero

(iv) large

(i) The _____ of a differential equation is the order of the highest derivative appearing in it.

(i) value

(ii) degree

(iii) dimension

(iv) order

(j) Lagrange's polynomial of degree two passes

(i) one point

(ii) two points

(iii) three points

(iv) infinite points

2. Answer any *three* of the following :

(a) Explain with examples exact and approximate numbers.

(b) Describe the method of false position.

(c) If the matrix A is such that

$$A = \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} (1 \quad 9 \quad 5)$$

then find $\det(A)$.

(d) Find x_3 for $f(x) = 0 \cdot 75x^3 - 2x^2 - 2x + 4$ using Newton-Raphson method. Assume that $x_0 = 2$.

(e) In the LU decomposition of the matrix

$$\begin{pmatrix} 2 & 2 \\ 4 & 9 \end{pmatrix}$$

if the diagonal elements of U are both 1, then find the lower diagonal entry l_{22} of L.

3. The velocity V km/min of a motorbike that starts from rest is given below :

t	2	4	6	8	10	12	14	16	18	20
V	10	18	25	29	32	20	11	5	2	0

Find the approximate distance covered in 20 minutes using Simpson's $\frac{1}{3}$rd rule.

4. Describe Lagrange's interpolation formula. Consider the following x_i's :

i	0	1	2
x_i	1	3	−2

Find $L_0(x)$, $L_1(x)$ and $L_2(x)$.

5. $f(x)$ is known and its values are given below. Here $f(x) = x^3 + 2x^2 + 3x + 1$:

x_i	0	1	2	3	4
f_i	1	7	23	55	109

Find $f(0 \cdot 5)$ and $f(1 \cdot 5)$ using Newton's forward difference formula.

6. Use Jacobi method to find the solution of the following set of linear equations :

$$5x_1 - 2x_2 + 3x_3 = -1$$
$$-3x_1 + 9x_2 + x_3 = 2$$
$$2x_1 - x_2 - 7x_3 = 3$$

(Turn Over

7. Solve by Runge-Kutta method for the DE

$$\frac{du}{dx} = -2u + x + 4, \quad u(0) = 1$$

to obtain $u(0 \cdot 2)$, using $\Delta x = 0 \cdot 2$.

* * *

Code : 303202

BCA 2nd Semester Theory Examination, 2017

Mathematics (Numerical Techniques)

Time : 3 hours *Full Marks : 60*

Instructions :

(i) There are seven questions in this Paper.

(ii) Attempt five questions in all.

(iii) Question No. 1 & 2 is compulsory.

(iv) The questions are of equal value.

1. Choose the correct answer (any six)

(a) In general the ratio of truncation error to that of round off error is

(i) 2 : 1

(ii) 1 : 1

(iii) 1 : 2

(iv) 1 : 3

(b) Rounding off the number 32.68673 to 4 significant digits, we get a number ....

(i) 32.68

(ii) 32.69

(iii) 32.67

(iv) 32.686

P.T.O.

https://www.akubihar.com

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(c) Every algebraic equation of the nth degree has exact .... roots.

(d) Jacobi's method is also known as

(i) Displacement method

(ii) Simultaneous displacement method

(iii) Simultaneous method

(iv) Diagonal method

(e) In bisection method if roots lies between a and b then f(a) × f(b) is ....

(i)	< 0	(ii)	$= 0$
(iii)	> 0	(iv)	none of these

(f) The technique for computing the value of the function inside the given argument is called ....

(i) interpolation

(ii) extrapolation

(iii) partial fraction

(iv) inverse interpolation

(g) If $\Delta f(x) = f(x+h) - f(x)$, then a constant k, Δk equ

(i)	1	(ii)	0
(iii)	$f(k)-f(0)$	(iv)	$f(x+k)-f(x)$

(v) None of these

(h) The root of the equation $x^3 - 2x - 5 = 0$ lies betw ....

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(i) 0 and 1 (ii) 1 and 2

(iii) 2 and 3 (iv) 3 and 4

(i) The order of errors the Simpson's rule for numerical integration with a step size h is

(i) h (ii) h^2

(iii) h^3 (iv) h^4

(j) If X is the true value of the quantity and X_1 is the approximate value then the relative error is E_R=..... and percentage error is E_p=------

2. Answer any three of the following:

(a) Explain the Bisection method for finding the real root of an equation $f(x)=0$.

(b) Show that the Newton method for finding reciprocals by solving $(1/x)-c=0$ results in the iteration: $x_{n+1} = x_n(2-cx_n), n \geq 0$

(c) Prove hat $(1+\Delta)(1-\nabla)=1$

(d) The function f(x) has the exact values shown in Table.

x	0	5	10	15	20	25
y = f(x)	6	10	13	17	23	21

Using Newton's forward difference interpolation method estimate the value of f(30).

(e) Find the absolute and the relative error when x=3.162 is used as an approximation to $x=\sqrt{10}$

P.T.O.

Code : 303202

3. The temperature T(in°c) and length l (in mm) of a heated ro[d] is given in the table. If l=a+bt, then find the best values of a and b.

T	20°	30°	40°	50°	60°	70°
I	600.1	600.4	600.6	600.7	600.9	601.[illegible]

4. Evaluate the integral $\int_0^{\pi/4} \sin 4x\,dx$, using Trapezoidal rule w[ith] n=4. Estimate the error bound and compute with exact err[or].

5. Estimate the missing term in the following data.

x	0	1	2	3	4
y	1	3	9	....	81

6. Consider the linear system:

$0.5x_1+1.1x_2+3.1x_3=6.0$

$2.0x_1+4.5x_2+0.4x_3=0.02$

$5.0x_1+1.0x_2+6.5x_3=1.0$

Solve the system using Gauss elimination method.

7 Why Runge-Kutta method is more effective in finding solution of differential equation $y'=f(x,y)$ w[ith] $y(x_0)=y_0$? Explain Runge-Kutta second order form[ula]. For $y'=y+x$ with $y(0)=1$ and h-=0.1, find the valu[e] K_1 in fourth order method.

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Code: 303202

Code : 303202

BCA 2nd Semester Exam., 2018

MATHEMATICS (Numerical Techniques)

Time : 3 hours *Full Marks* : 60

Instructions :

(i) All questions carry equal marks.

*(ii) There are **SEVEN** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question Nos. **1** & **2** are compulsory.*

1. Choose the correct answer of the following (any *six*) :

(a) The number of significant digits in the number 204·020050 is

(i) 5

(ii) 6

(iii) 8

(iv) 9

(Turn Over)

(b) The convergence of which of the following methods is sensitive to starting value?

(i) False position

(ii) Gauss-Seidel method

(iii) Newton-Raphson method

(iv) All of the above

(c) Newton-Raphson method is used to find the root of the equation $x^2 - 2 = 0$. If iterations are started from -1, then iterations will

(i) converge to -1

(ii) converge to $\sqrt{2}$

(iii) converge to $-\sqrt{2}$

(iv) No convergence

(d) In the Gauss elimination method for solving a system of linear algebraic equations triangularization leads to

(i) diagonal matrix

(ii) lower triangular matrix

(iii) upper triangular matrix

(iv) singular matrix

(Continued)

(e) In which of the following methods, we approximate the curve of solution by the tangent in each interval?

(i) Picard's method

(ii) Euler's method

(iii) Newton's method

(iv) Runge-Kutta method

(f) Match the following :

A.	Newton-Raphson	1.	Integration
B.	Runge-Kutta	2.	Root finding
C.	Gauss-Seidel	3.	Ordinary differential equations
D.	Simpson's rule	4.	Solution of system of linear equations

The correct matching is

(i) A–2, B–3, C–4, D–1

(ii) A–3, B–2, C–1, D–4

(iii) A–1, B–4, C–2, D–3

(iv) A–4, B–1, C–2, D–3

(g) If $\Delta f(x) = f(x+h) - f(x)$, then a constant k, Δk equals

(i) 1

(ii) 0

(iii) $f(k) - f(0)$

(iv) $f(x+k) - f(x)$

(v) None of the above

(Turn Over)

(h) A root of the equation $x^3 - x - 11 = 0$ correc to four decimals using bisection method is

(i) 2·3737

(ii) 2·3838

(iii) 2·3736

(iv) None of the above

(i) The order of errors in the Simpson's rule fo numerical integration with a step size h is

(i) h

(ii) h^2

(iii) h^3

(iv) h^4

(j) In which of the following methods prop choice of initial value is very important?

(i) Bisection method

(ii) False position

(iii) Newton-Raphson method

(iv) None of the above

(Continue

2. Answer any *three* of the following :

(a) Use quadratic convergent method to find the first approximation $x^{(1)}$ of $\sqrt[3]{28}$ if $x^{(0)} = 3$.

(b) Show that the Newton method for finding reciprocals by solving $(1/x) - c = 0$ results in the iteration, $x_{n+1} = x_n(2 - cx_n)$, $n \geq 0$.

(c) Let $g(x)$ be a continuous function on $[a, b]$, and suppose that g satisfies the property $a \leq x \leq b$ implies that $a \leq g(x) \leq b$. Then the equation $x = g(x)$ has at least one solution in $[a, b]$.

(d) The function $f(x)$ has the exact values shown in the table below :

x	1	3	5
$f(x)$	4	−2	10

Using Newton's forward difference interpolation method, estimate the value of $f(6)$.

(e) Find the absolute and the relative error when $x = 3 \cdot 162$ is used as an approximation to $x = \sqrt{10}$.

3. Show that equation $xe^x - 1$ has a root in $[1/2, 1]$ and find the approximation to this root within 10^{-1}, using bisection method.

(Turn Over)

4. Evaluate the integral $\int_0^{\pi/4} \sin 4x\, dx$, using trapezoidal rule with $n = 4$. Estimate the error bound and compute with exact error.

5. Given the table of values :

x	1·0	1·05	1·08	1·1
$f(x)$	2·72	3·29	3·66	3·90

Construct the best quadratic Lagrange interpolation polynomial to approximate the equation, $f(x) = 3xe^x - 2e^x$ at $x = 1 \cdot 04$.

6. Consider the linear systems :

$$\begin{aligned} 0 \cdot 5x_1 + 1 \cdot 1x_2 + 3 \cdot 1x_3 &= 6 \cdot 0 \\ 2 \cdot 0x_1 + 4 \cdot 5x_2 + 0 \cdot 4x_3 &= 0 \cdot 02 \\ 5 \cdot 0x_1 + 1 \cdot 0x_2 + 6 \cdot 5x_3 &= 1 \cdot 0 \end{aligned}$$

Solve the systems using Gauss elimination method.

7. Use Runge-Kutta of order 2 with $h = 0 \cdot 1$ to find the approximate value of $y(0 \cdot 3)$ of the initial value problem, $y' = y(2 - y)$; $y(0) = 0 \cdot 1$.

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Code : 303202

BCA 2nd Semester Exam., 2019

MATHEMATICS (Numerical Technique)

Time : 3 hours | *Full Marks* : 60

Instructions :

(i) The marks are indicated in the right-hand margin.

(ii) There are ***SEVEN*** *questions in this paper.*

(iii) Attempt ***FIVE*** *questions in all.*

(iv) Question Nos. ***1*** *&* ***2*** *are compulsory.*

1. Answer any *six* of the following : $2 \times 6 = 12$

   (a) Round off the number 865250 to four significant figures and compute absolute error E_a, relative error E_r and percentage error E_p.

   (b) Solve the equation $x^3 - 7x^2 + 36 = 0$. Given that one root is double of another.

   (c) Evaluate $\Delta(\tan^{-1} x)$.

   (d) If $\Delta f(x) = f(x+h) - f(x)$, then find the value of $\Delta(k)$ where k is constant.

   (e) If $y(x) = x(x+1)(x+2)(x+3)$, then find $\Delta y(x)$ taking $h = 1$.

   (f) Construct Newton's divided difference table for the following data :

X	5	7	11	13	17
Y	150	392	1452	2366	5202

   (g) Construct Newton's forward interpolation table for the following data :

X	4	6	8	10
Y	1	3	8	16

   (h) Construct Newton's backward interpolation table for the following data :

X	80	85	90	95	100
Y	5026	5674	6362	7088	7854

   (i) Find the condition that the cubic $x^3 - lx^2 + mx - n = 0$ should have its roots in arithmetic progression.

   (j) Write numerical expression for Simpson's $1/3$ and Simpson's $3/8$ rule.

2. Answer any *three* of the following : 4×3=12

(a) Evaluate :

$$\Delta^2\left(\frac{5x+12}{x^2+5x+16}\right)$$

(b) Find the missing term in the following table :

X	2	3	4	5	6
Y	45·0	49·2	54·1	—	67·4

(c) Evaluate :

$$\int_0^1 \frac{dx}{1+x^2}$$

Hence obtain the approximate value of π.

(d) Find a root of the equation $x^3-4x-9=0$ using the bisection method correct to three decimal places.

(e) Given the values :

X	5	7	11	13	17
Y	150	392	1452	2366	5202

Evaluate $f(9)$ using Lagrange's interpolation formulae.

3. Find the cubic polynomial which takes the following values :

X	0	1	2	3
F(x)	1	2	1	10

Hence or otherwise evaluate $f(4)$. 12

4. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using—

(i) trapezoidal rule;

(ii) Simpson's 1/3 rule;

(iii) Simpson's 3/8 rule.

5. Apply Gauss elimination method to solve the equations $x+4y-z=-5$; $x+y-6z=-12$; $3x-y-z=4$.

6. Apply Runge-Kutta fourth-order method to find an approximate value of y when $x=0{\cdot}2$. Given that $dy/dx = x+y$ and $y=1$ when $x=0$.

7. Find the number of men getting wages between ₹ 10 and ₹ 15 from the following table :

Wages (in ₹)	0–10	10–20	20–30	30–40
Frequency	9	30	35	42

★★★

Code : 303202

BCA 2nd Semester Exam., 2021

MATHEMATICS

(Numerical Techniques)

Time : 3 hours *Full Marks* : 60

Instructions :

(i) *The marks are indicated in the right-hand margin.*

(ii) *There are **SEVEN** questions in this paper.*

(iii) *Attempt **FIVE** questions in all.*

(iv) *Question Nos. **1** and **2** are compulsory.*

1. Choose the correct answer of the following (any *six*) : 2×6=12

(a) Match the following :

A. Newton-Raphson	1. Integration
B. Runge-Kutta	2. Root finding
C. Gauss-Seidel	3. Ordinary differential equations
D. Simpson's rule	4. Solution of system of linear equations

The correct sequence is

(i) A B C D
2 3 4 1

(ii) A B C D
3 2 1 4

(iii) A B C D
1 4 2 3

(iv) A B C D
4 1 2 3

(b) If $\Delta f(x) = f(x+h) - f(x)$, then for a constant k, Δk equal

(i) 1

(ii) 0

(iii) $f(k) - f(0)$

(iv) $f(x+k) - f(x)$

(c) An equation involving only a linear polynomial is called a

(i) linear equation

(ii) quadratic equation

(iii) simultaneous equation

(iv) None of the above

(d) The pairs of equations $x + 2y - 5 = 0$ and $-4x - 8y + 20 = 0$ have

(i) unique solution

(ii) exactly two solutions

(iii) infinitely many solutions

(iv) no solution

(e) If x and y are real numbers, then find $\max(x, y) + \min(x, y)$.

(i) x

(ii) y

(iii) x/y

(iv) None of the above

(f) If $\alpha_1, \alpha_2, \ldots, \alpha_n$ are n roots of the equation $x^n - 1 = 0$, then

$$(1-\alpha_1)(1-\alpha_2)\cdots(1-\alpha_{n-1})$$

is

(i) 0

(ii) 1

(iii) $(n-1)$

(iv) n

(g) Let $A = [-2]$. Then $\det(A) = |A|$ is

(i) 2

(ii) 1

(iii) 0

(iv) None of the above

(h) Which of the following is an assumption of Jacobi's method?

(i) The coefficient matrix has no zeros on its main diagonal

(ii) The rate of convergence is quite slow compared with other methods

(iii) Iteration involved in Jacobi's method converges

(iv) The coefficient matrix has zeros on its main diagonal

(i) Which of the following is a direct method?

(i) Gauss elimination method

(ii) Newton-Raphson method

(iii) Regula-Falsi method

(iv) Jacobi method

(j) Interpolation helps us in estimating

(i) missing values of a series

(ii) an intermediate value

(iii) the value for a given entry

(iv) All of the above

2. Answer any *three* of the following questions : $4 \times 3 = 12$

(a) One kind of bacteria y grows according to the equation $dy/dt = ky$, where k is a constant and t is measured in years. If the amount of the bacteria doubles every 5 days, then find the value of k.

(b) Find the absolute error and the relative error when $x = 3 \cdot 162$ is used as an approximation to $x = \sqrt{10}$.

(c) Solve the system of linear equations

$$2x + y + z = 10$$
$$3x + 2y + 3z = 18$$
$$x + 4y + 9z = 16$$

by Gauss elimination method.

(d) Using Euler's method, solve the following differential equation $y' = -y$ subject to $y(0) = 1$.

(e) Derive the formula to find the second-order derivative using the forward differences.

3. Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$ 12

4. Compute the real root of $x \log_{10} x - 1 \cdot 2 = 0$ correct to five decimal places by Regula-Falsi method. 12

5. Find $\sqrt{16}$ to two places of decimal by Newton-Raphson method. 12

6. Write down the trapezoidal rule in numerical integration and then solve the following using this technique : 12

$$\int_0^6 \frac{1}{1+x^2} dx$$

7. A rocket is launched from the ground. Its acceleration is registered during the first 80 seconds and is in the table below. Using Simpson's 1/3rd rule, find the velocity of the rocket at $t = 80$ sec. 12

t (sec) :	0	10	20	30	40	50	60	70	80
f (cm/sec^2) :	30	31·63	33·34	35·47	37·75	40·33	43·25	46·69	40·67

★★★

Code : 303202

BCA 2nd Semester Exam., 2022

MATHEMATICS

(Numerical Techniques)

Time : 3 hours *Full Marks* : 60

Instructions :

(i) *The marks are indicated in the right-hand margin.*

(ii) *There are* **SEVEN** *questions in this paper.*

(iii) *Attempt* **FIVE** *questions in all.*

(iv) *Question Nos.* **1** *and* **2** *are compulsory.*

1. Answer any *six* of the following as directed : $2 \times 6 = 12$

(a) What is the order of the truncation error of the trapezoidal rule as function of n, the number of trapezoids?

(b) The numerical techniques more commonly involve

(i) iterative method

(ii) direct method

(iii) elimination method

(iv) reduction method

(Choose the correct option)

(c) The rate of convergence of Newton-Raphson method is generally

(i) linear

(ii) quadratic

(iii) superlinear

(iv) cubic

(Choose the correct option)

(d) Find the number of significant digits in the number 219900.

(e) The equation $f(x)$ is given as $x^3 - x^2 + 4x - 4 = 0$. Considering the initial approximation at $x = 2$, the value of next approximation correct up to two decimal places is given as

(i) 0·67

(ii) 1·33

(iii) 1·00

(iv) 1·50

(Choose the correct option)

(f) Which of the following is an iterative method?

(i) Jacobi method

(ii) Gauss elimination method

(iii) Gauss-Seidel method

(iv) Factorization method

(Choose the correct option)

(g) In an iterative method, the amount of computation depends on the

(i) number of variables

(ii) degree of accuracy

(iii) rounding of errors

(iv) ease of using the operators

(Choose the correct option)

(h) In which of the following methods, we approximate the curve of solution by the tangent in each interval?

(i) Picard's method

(ii) Euler's method

(iii) Newton's method

(iv) Runge-Kutta method

(Choose the correct option)

(i) Which of the following is a direct method?

(i) Gauss elimination method

(ii) Newton-Raphson method

(iii) Regula Falsi method

(iv) Jacobi method

(Choose the correct option)

(j) Interpolation helps us in estimating

(i) the missing values of a series

(ii) an intermediate value

(iii) the value for a given entry

(iv) All of the above

(Choose the correct option)

2. Answer any *three* of the following : 4×3=12

(a) Approximate value of variable is 3·436. If true value X of variable lies in the interval [3·431, 3·441], then find the value of upper limit of the absolute error.

(b) Find sin(0·197) from the following table :

X	0·15	0·17	0·19	0·21	0·23
$\sin X$	0·14944	0·16918	0·18886	0·20846	0·22798

(c) Solve the system

$$\begin{aligned} 2x+y-z&=-1 \\ x-2y+3z&=9 \\ 3x-y+5z&=14 \end{aligned}$$

by Gauss elimination method.

(d) Explain the difference between Newton's method and Secant method.

(e) Find the number of significant digits in the number $0 \cdot 01850 \times 10^3$.

3. The velocity of a body is given by $v(t)$

$$\begin{aligned} &= 2t, && 1 \le t \le 5 \\ &= 5t^2+3, && 5 < t \le 14 \end{aligned}$$

where t is given in seconds, and v is given in m/s. Use the two-segment trapezoidal rule to find the distance covered by the body from $t=2$ to $t=9$ seconds. 12

4. Compute the real root of $x \log_{10} x - 1 \cdot 2 = 0$ correct to five decimal places by Regula Falsi method. 12

5. Given

$$\frac{d^2y}{dx^2} = 6x - 0 \cdot 5x^2, \; y(0)=0, \; y(12)=0$$

Using finite difference technique with step size 4, approximate the value of d^2y/dx^2 at $y(4)$. 12

6. Using iteration method, obtain approximate root correct up to four decimal places of $x=(5-x)^{1/3}$. 12

7. Evaluate the following integration using trapezoidal rule : 12

$$\int_{0.2}^{2.2} x e^{x} \, dx$$

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Code : 303202

BCA 2nd Semester Exam., 2023

MATHEMATICS

(Numerical Techniques)

Time : 3 hours *Full Marks* : 60

Instructions :

(i) The marks are indicated in the right-hand margin.

(ii) There are **SEVEN** *questions in this paper.*

(iii) Attempt **FIVE** *questions in all.*

(iv) Question Nos. **1** *and* **2** *are compulsory.*

1. Answer any *six* of the following as directed : 2×6=12

(a) What is required to perform chi-square test?

(b) The Newton-Raphson method is also called as ______.

(Fill in the blank)

(c) The equation $f(x)$ is given as $x^2 - 4 = 0$. Considering the initial approximation at $x = 6$, then the value of x_1 is given as ______.

(Fill in the blank)

(d) The points where the Newton-Raphson method fails are called ______.

(Fill in the blank)

(e) In the Gauss elimination method for solving a system of linear algebraic equations, triangularization leads to ______.

(Fill in the blank)

(f) If $\Delta f(x) = f(x+h) - f(x)$, then for a constant k, Δk equals ______.

(Fill in the blank)

(g) Following are the values of a function $y(x)$:

$$y(-1) = 5;\ y(0),\ y(1) = 8$$

dy/dx

at $x = 0$ as per Newton's central difference scheme is ______.

(Fill in the blank)

(h) How many zeroes does a polynomial of degree n have?

(i) Both Newton's method and the secant method are examples of fixed-point iterations.

(State True or False)

(j) If A is a 2×2 invertible matrix, then what is the inverse of $2A$?

2. Answer any *three* of the following : 4×3=12

(a) State Simpson's $\frac{1}{3}$rd rule with an example.

(b) Construct an algorithm to implement the method of false position.

(c) Define order of convergence of an iterative method for finding an approximation to the location of a root of $f(x) = 0$. Find the order of convergence of Newton's method.

(d) Explain how to perform detection of errors using difference table.

(e) State Runge-Kutta method of second-order with an example.

3. Verify that the function $f(x) = \sin x$ has a zero in the interval (3, 4). Perform three iterations of bisection method and verify that each approximation satisfies the theoretical error bound. The exact location of the zero is $P = \pi$. 12

4. Starting with initial vector $X^{(0)}$, perform three iterations of Gauss-Seidel method to

$$2x - y = -1$$
$$-x + 4y + 2z = 3$$
$$2y + 6z = 5$$

12

5. Describe the difference between Euler's method and modified Euler's method with an example. 12

6. Discuss the divided differences and their properties. 12

7. Describe Newton's formulae for interpolation with an example. 12

★ ★ ★

Code : 303202

BCA 2nd Semester Exam., 2025

MATHEMATICS

(Numerical Techniques)

Time : 3 hours *Full Marks* : 60

Instructions :

(i) The marks are indicated in the right-hand margin.

(ii) There are **SEVEN** *questions in this paper.*

(iii) Attempt **FIVE** *questions in all.*

(iv) Question Nos. **1** *and* **2** *are compulsory.*

1. Choose the correct answer (any *six*) : $2 \times 6 = 12$

(a) The number of significant digits in the number 204·020050 is

(i) 5

(ii) 6

(iii) 8

(iv) 9

(b) Jacobi's method is also known as

(i) displacement method

(ii) simultaneous displacement method

(iii) simultaneous method

(iv) diagonal method

(c) The convergence of which of the following methods is sensitive to starting value?

(i) False position method

(ii) Gauss-Seidel method

(iii) Newton-Raphson method

(iv) All of the above

(d) If $\Delta f(x) = f(x+h) - f(x)$, then a constant k, Δk equals

(i) 1

(ii) 0

(iii) $f(k) - f(0)$

(iv) $f(x+k) - f(x)$

(e) The root of $x^3 - 2x - 5 = 0$ correct to three decimal places by using Newton-Raphson method is

(i) 2·0946

(ii) 1·0404

(iii) 1·7321

(iv) 0·7011

(f) Errors may occur in performing numerical computation on the computer due to

(i) rounding errors

(ii) power fluctuation

(iii) operator fatigue

(iv) All of the above

(g) In the Gauss elimination method for solving a system of linear algebraic equations, triangularization leads to

(i) diagonal matrix

(ii) lower triangular matrix

(iii) upper triangular matrix

(iv) singular matrix

(h) In which of the following methods, we approximate the curve of solution by the tangent in each interval?

(i) Picard's method

(ii) Euler's method

(iii) Newton's method

(iv) Runge-Kutta method

(i) Which of the following is a direct method?

(i) Gauss elimination method

(ii) Newton-Raphson method

(iii) Regula-Falsi method

(iv) Jacobi method

(j) Interpolation helps us in estimating

(i) missing values of a series

(ii) an intermediate value

(iii) the value for a given entry

(iv) All of the above

2. Answer any *three* of the following : $4\times3=12$

(a) The population of a town were as given in the table below :

Year	1921	1931	1941	1951	1961
Population (in thousands)	46	66	81	93	101

Estimate the population for the year 1955.

(b) Find y, when $x = 8$ for

x	0	5	10	15	20	25
y	7	11	14	18	24	32

(c) Solve the system

$$5x - 2y + z = 4$$
$$7x + y - 5z = 8$$
$$3x + 7y + 4z = 10$$

by Gauss elimination method.

(d) Find the number of positive and negative roots of the equation $x^4 + 4x^3 - 4x - 13 = 0$, using Descartes' rule.

(e) Find the relative error of the number 1·53364.

3. Solve $y^n = (x^2 + y^2)(1 + y^2)$ for $x = 0 \cdot 5$ and $x = 1 \cdot 0$ by using Runge-Kutta method, with $x = 0,\ y = 1,\ y' = 0.$ 12

4. Compute the real root of $x \log_{10} x - 1 \cdot 2 = 0$ correct to five decimal places by Regula-Falsi method. 12

5. Find $\sqrt{12}$ to five places of decimal by Newton-Raphson method. 12

6. Evaluate

$$\int_0^6 \frac{1}{1 + x^2}\, dx$$

by using Simpson's $\frac{1}{3}$rd rule. 12

7. Solve the following system by iteration method : 12

$$27x + 6y - z = 85$$
$$6x + 15y + 2z = 72$$
$$x + y + 54z = 110$$

★★★